

Network Flow

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Max Flow: Problem Formulation

Input: A flow network $G = (V, E, c)$, $c : E \rightarrow \mathbb{R}^+$

$s \in V$ a source and $t \in V$ a sink

$f : E \rightarrow \mathbb{R}^+$ ($f_e = f(e)$) is a flow if it satisfies

1 **capacity constraints** $\forall e \in E : 0 \leq f_e \leq c_e$

2 **flow conservation constraints** $\forall v \in V, v \neq s, t : f^{out}(v) = f^{in}(v)$

$$size(f) = f^{out}(s) = f^{in}(t)$$

Output: A flow f of maximum size

Let f be a flow in G and let $[A, \bar{A}]$ be any $s - t$ cut in G , then

$$size(f) \leq c([A, \bar{A}])$$

Max Flow: Algorithm

There is no known divide-and-conquer or dynamic programming algorithm for max-flow

We try a greedy strategy – build up flow little bit at a time

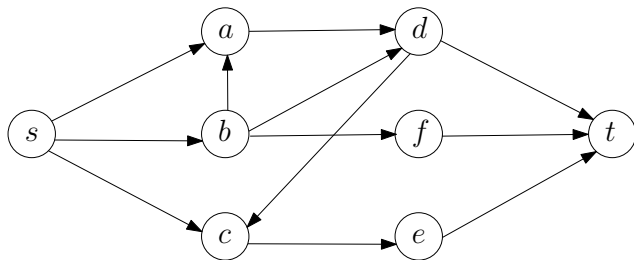
- 1 Start with a 0 flow

▷ Note: the flow f with $f_e = 0$ for all $e \in E$ satisfies both constraints

- 2 Add more flow to f via a $s - t$ path

Max Flow: Algorithm

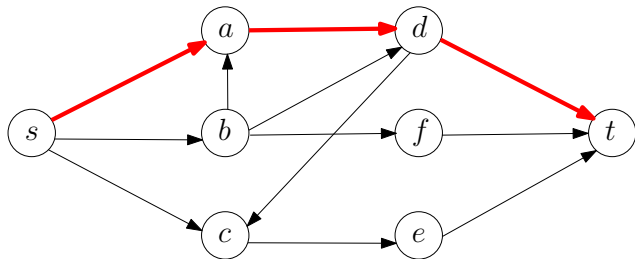
Consider the flow network – all edge capacities are 1



Max Flow: Algorithm

Consider the flow network – all edge capacities are 1

Add flow of size 1 via the $s - t$ path s, a, d, t

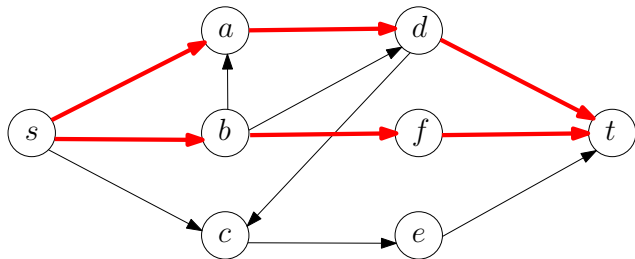


Max Flow: Algorithm

Consider the flow network – all edge capacities are 1

Add flow of size 1 via the $s - t$ path s, a, d, t

Add flow of size 1 via the $s - t$ path s, b, f, t



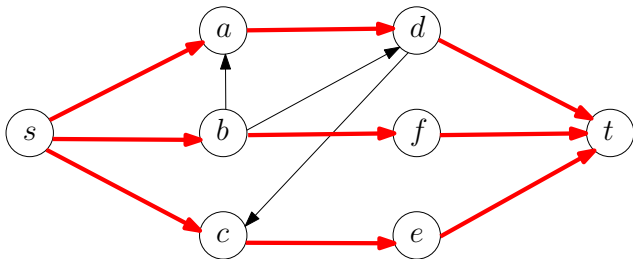
Max Flow: Algorithm

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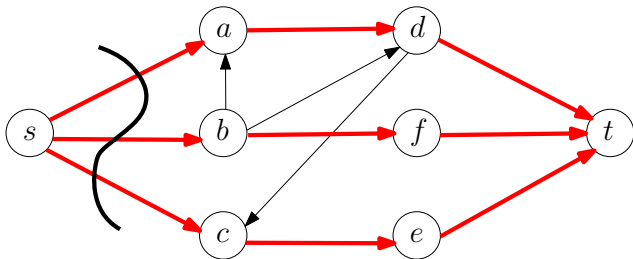
Max Flow: Algorithm

Consider the flow network – all edge capacities are 1

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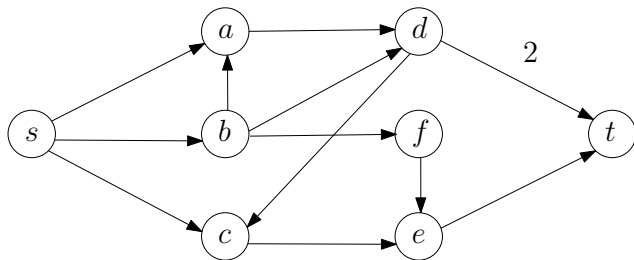
Add flow of size 1 via the $s - t$ path s, c, e, t



There is an $s - t$ cut of capacity 3, hence this flow is maximum possible

Max Flow: Algorithm

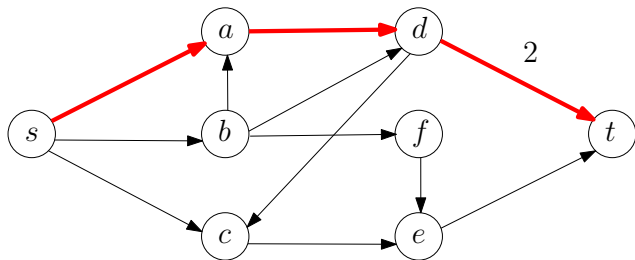
Consider the flow network – with not all capacities = 1



Max Flow: Algorithm

Consider the flow network – with not all capacities = 1

Add flow of size 1 = path-bottleneck via the $s - t$ path s, a, d, t

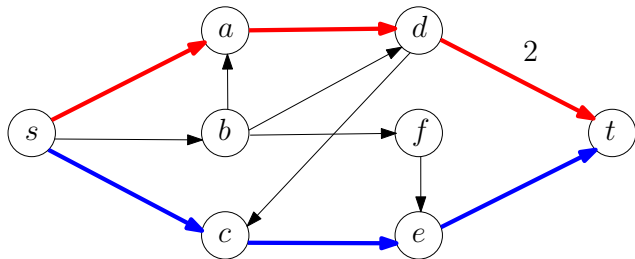


Max Flow: Algorithm

Consider the flow network – with not all capacities = 1

Add flow of size 1 = path-bottleneck via the $s - t$ path s, a, d, t

Add flow of size 1 via the $s - t$ path s, b, f, t



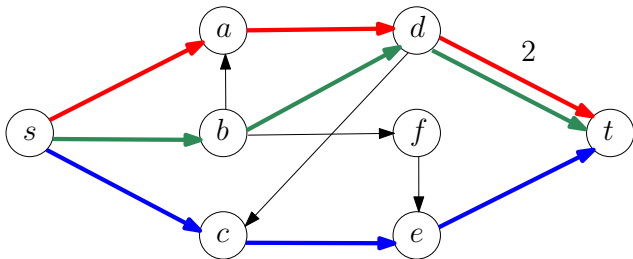
Max Flow: Algorithm

Consider the flow network – with not all capacities = 1

Add flow of size 1 = path-bottleneck via the $s - t$ path s, a, d, t

Add flow of size 1 via the $s - t$ path s, b, f, t

Add flow of size 1 via the $s - t$ path s, c, e, t



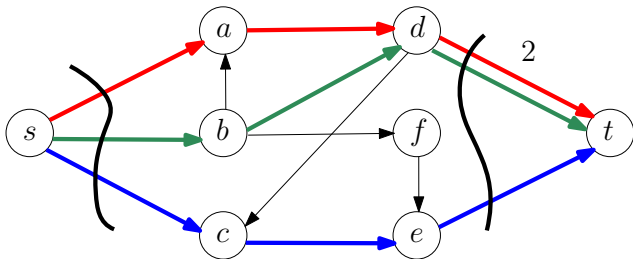
Max Flow: Algorithm

Consider the flow network – with not all capacities = 1

Add flow of size 1 = **path-bottleneck** via the $s - t$ path s, a, d, t

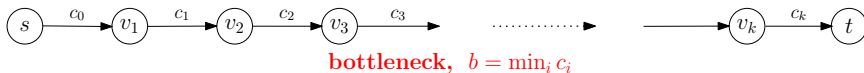
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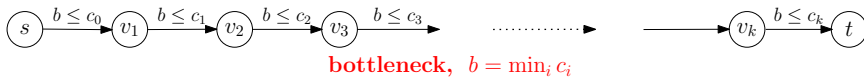


There is an $s - t$ cut of capacity 3, hence this flow is maximum possible

Max Flow: Adding Flow along a path



Adding flow along a path ensures that flow conservation constraints remain satisfied



Adding a flow along a path equal to the bottleneck of the path doesn't violate the capacity constraints

These two facts ensure that in any intermediate step of such a greedy algorithm the flow indeed is a valid flow